Biosynthesis

- Formation of a chemical compound by a living organism.
- The pathways for generally modifying and synthesizing carbohydrates, proteins, fats, and nucleic acids are found to be essentially the same in all organisms, apart from minor variations.
- These processes are collectively described as **primary metabolism,** with the compounds involved in the pathways being termed primary metabolites**.**

PRIMARY METABOLISM

- Primary metabolism comprises the chemical processes that every plant must carry out every day in order to survive and reproduce its line.
  - Photosynthesis
  - Glycolysis
  - Citric Acid Cycle (Krebs cycle)
  - Synthesis of amino acids
  - Transamination
  - Synthesis of proteins and enzymes
  - Synthesis of coenzymes
  - Synthesis of structural materials
  - Duplication of genetic material
  - Reproduction of cells (growth)
  - Absorption of nutrients

SECONDARY METABOLISM

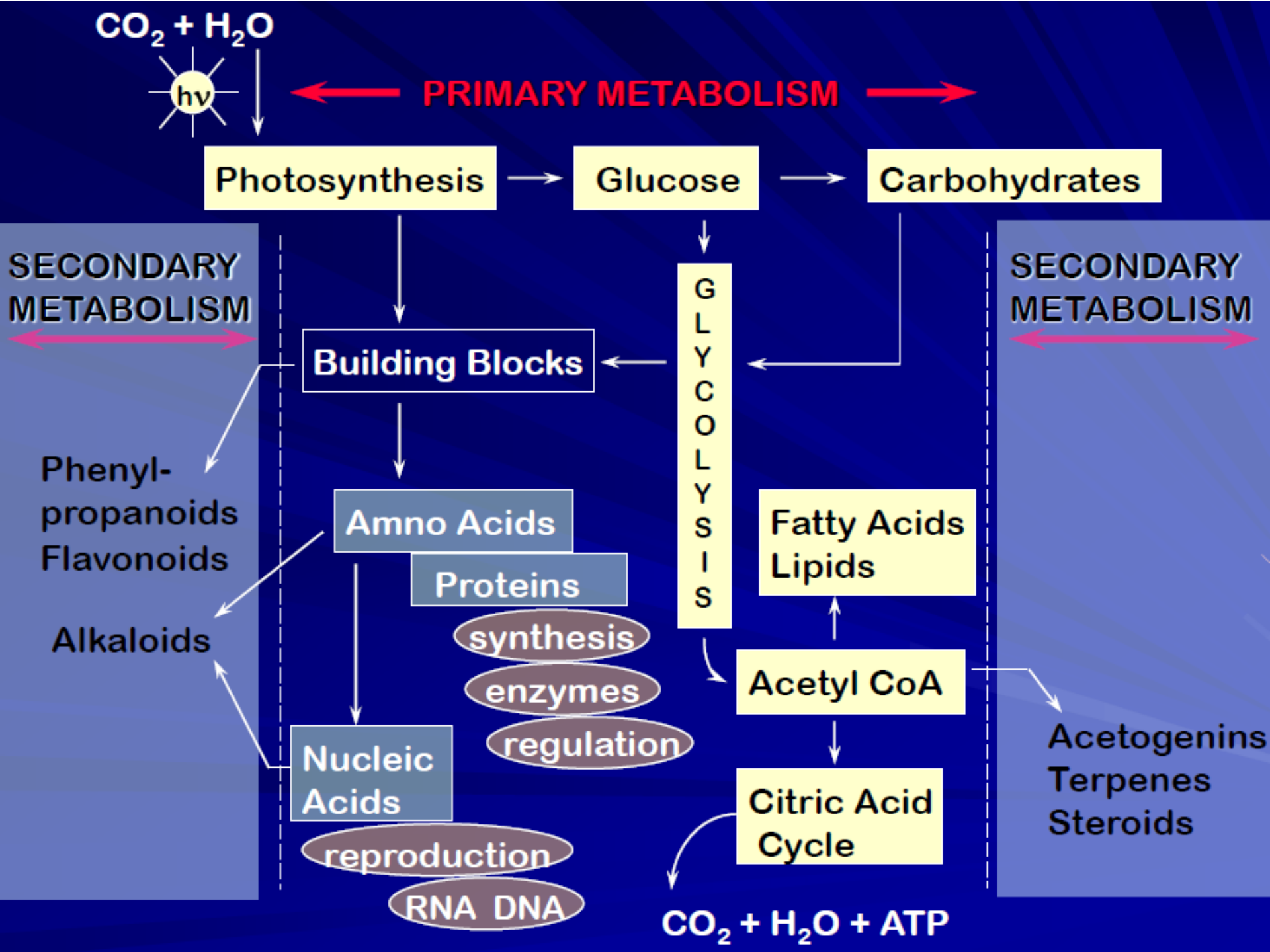
- **Secondary metabolism**, metabolic pathways that are not essential for growth, development or reproduction**.**
- **Secondary metabolites** are those **chemical compounds** in organisms that are not directly involved in the normal growth, development or reproduction of an organism. In this sense they are **"secondary"**.
- Secondary metabolites, are found in only specific organisms, or groups of organisms, and are an expression of the individuality of species.
- Every type of secondary metabolite is characteristic for each organism or group of organism
  - Natural products chemistry (Alkaloids, glycosides, essential oils,…).

Primary and Secondary Plant metabolites

- The chemical constituents found in plants are often classified as either primary or secondary metabolites.
- In all living beings, chemical compounds are synthesized and broken down via a series of chemical reactions, each mediated by an enzyme.
- This complex of processes is known as metabolism, which comprises catabolism(breakdown) and anabolism (synthesis).

Primary and Secondary Plant metabolites

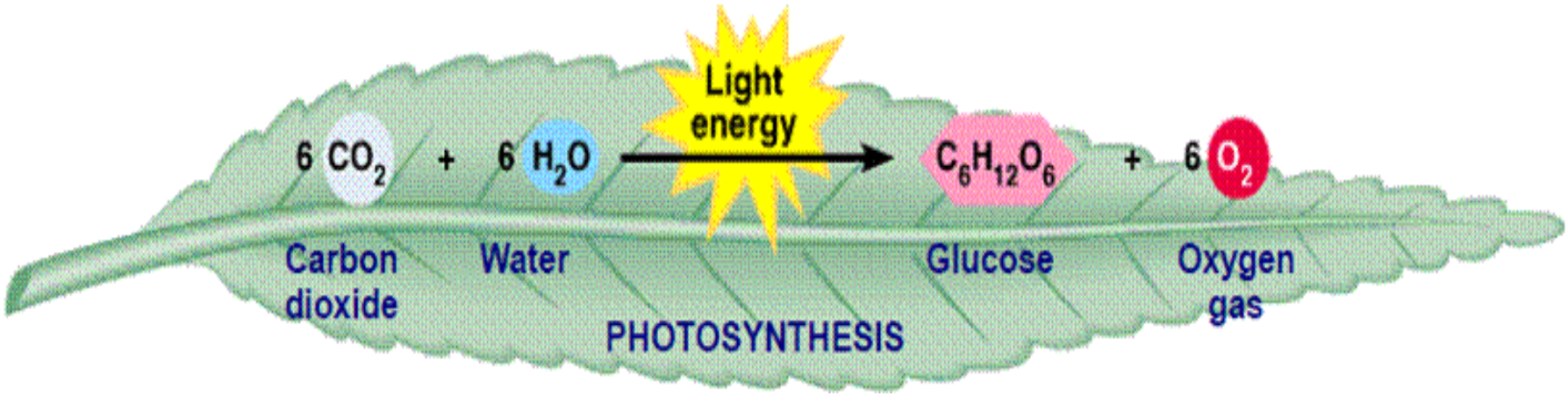
- Metabolites are organic compounds synthesized by organisms using enzyme-mediated chemical reactions called metabolic pathways.
- Primary metabolites have functions that are essential to growth and development and are therefore present in all plants.
- In contrast, secondary metabolites are variously distributed in the plant kingdom, and their functions are specific to the plants in which they are found.


$CO_2 + H_2O$
hν
PRIMARY METABOLISM
Photosynthesis
Glucose
Carbohydrates
SECONDARY METABOLISM
SECONDARY METABOLISM
GLYCOLYSIS
Building Blocks
Phenyl-propanoids Flavonoids
Alkaloids
Amno Acids
Proteins
synthesis
enzymes
regulation
Nucleic Acids
reproduction
RNA DNA
Fatty Acids Lipids
Acetyl CoA
Citric Acid Cycle
$CO_2 + H_2O + ATP$
Acetogenins Terpenes Steroids

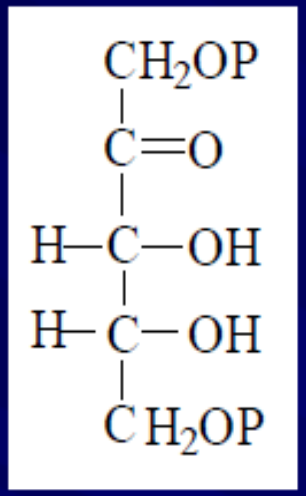
Primary Plant metabolites

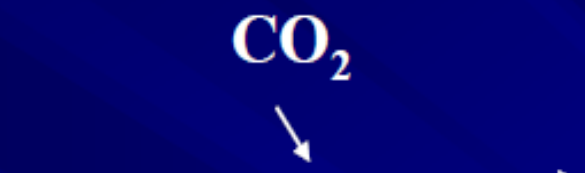
- All organisms posses similar metabolic pathways for the synthesis and use of certain essential chemicals; sugars, amino acids, common fatty acids, nucleotides, and the polymers derived from them(including polysaccharides, proteins, lipids, RNA and DNA).
- This is primary metabolism, and these compounds, which are essential for the survival and well being of the organism, are primary metabolites.
- In plants such compounds are responsible for the primary life processes of respiration, photosynthesis, growth, development and other essential functions.

- Photosynthesis is the process by which autotrophic organisms use light energy to make sugar and oxygen gas from carbon dioxide and water.
- Almost all plants are photosynthetic autotrophs, as some bacteria and protists.

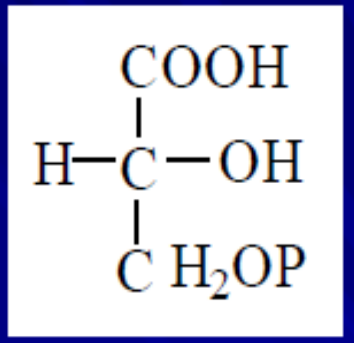


Calvin Cycle



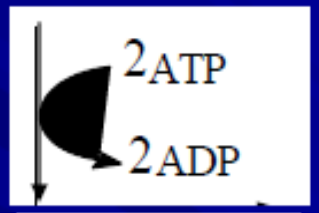


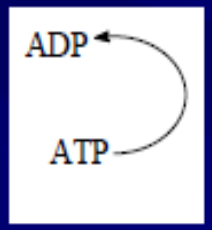
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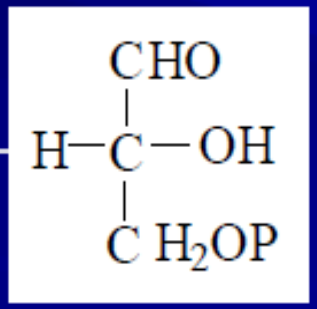
Ribulose 1,5bisPO4 (RuBP)

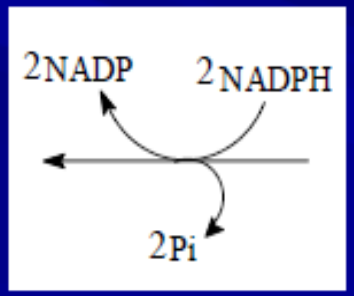
3-Phosphoglyceric acid (PGA)



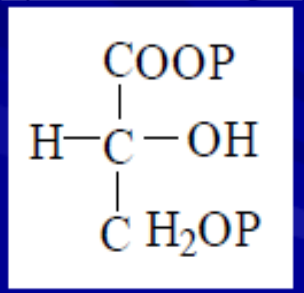


2





2



Glucose

Glyceraldehyde-3- phosphate (G3P)

1,3 diphosphoglycerate

The Calvin Cycle in Photosynthesis

- The Calvin cycle is the mechanism for turning CO_2 into glucose.
- The cycle fixes 6 CO_2 to 6 ribulose 1,5-bisPO4 acceptor molecules, makes glucose with 6 of the carbons, and returning the rest to continue the cycle.
- 6 CO_2 $\rightarrow$ 1 $C_6H_{12}O_6$ (6C)
- 18 ATP + 12 NADPH $\rightarrow$ 1 $C_6H_{12}O_6$

Glycolysis

Stage 1

Glucose

Hexokinase — ATP / ADP

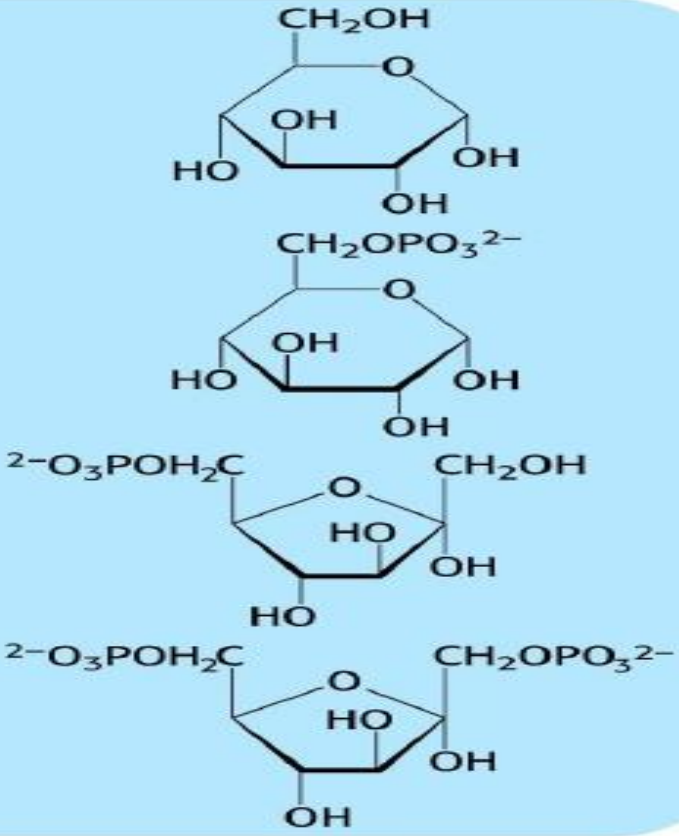
Glucose-6-phosphate

Phosphoglucose isomerase

Fructose-6-phosphate

Phospho-fructokinase — ATP / ADP

Fructose-1,6-bisphosphate



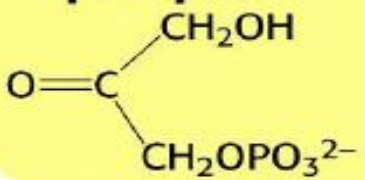
Stage 2

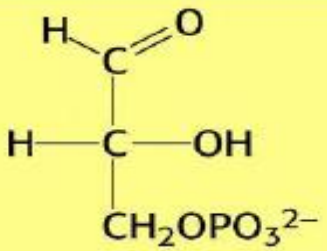
Triose phosphate isomerase

Aldolase

Dihydroxyacetone phosphate

Glyceraldehyde 3-phosphate





Stage 3

2 X

Glyceraldehyde 3-phosphate dehydrogenase

P_i, NAD^+

NADH

1,3-Bisphosphoglycerate

Phosphoglycerate kinase

ADP

ATP

3-Phosphoglycerate

Phosphoglycerate mutase

2-Phosphoglycerate

Enolase

H_2O

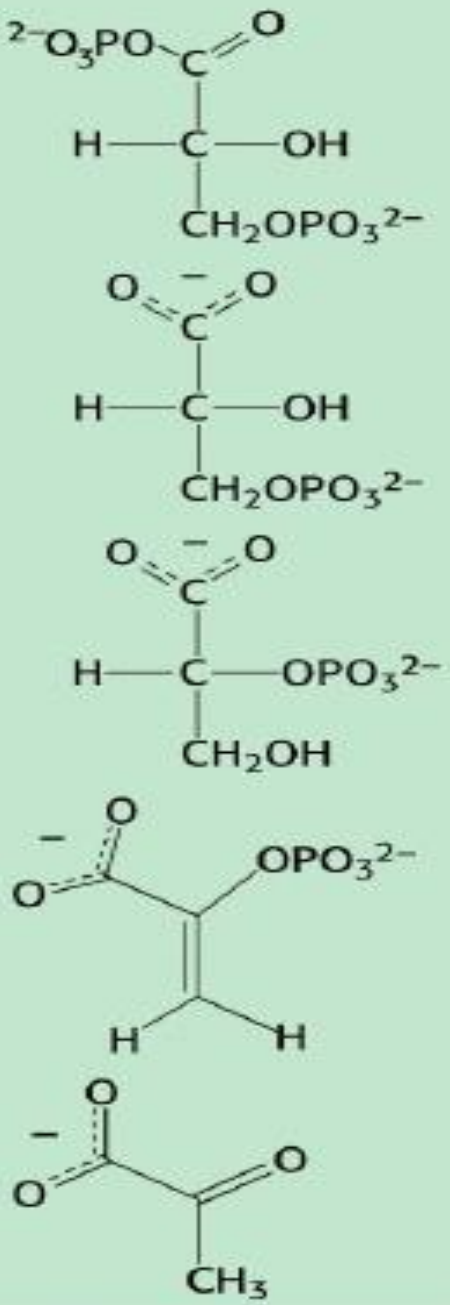
Phosphoenolpyruvate

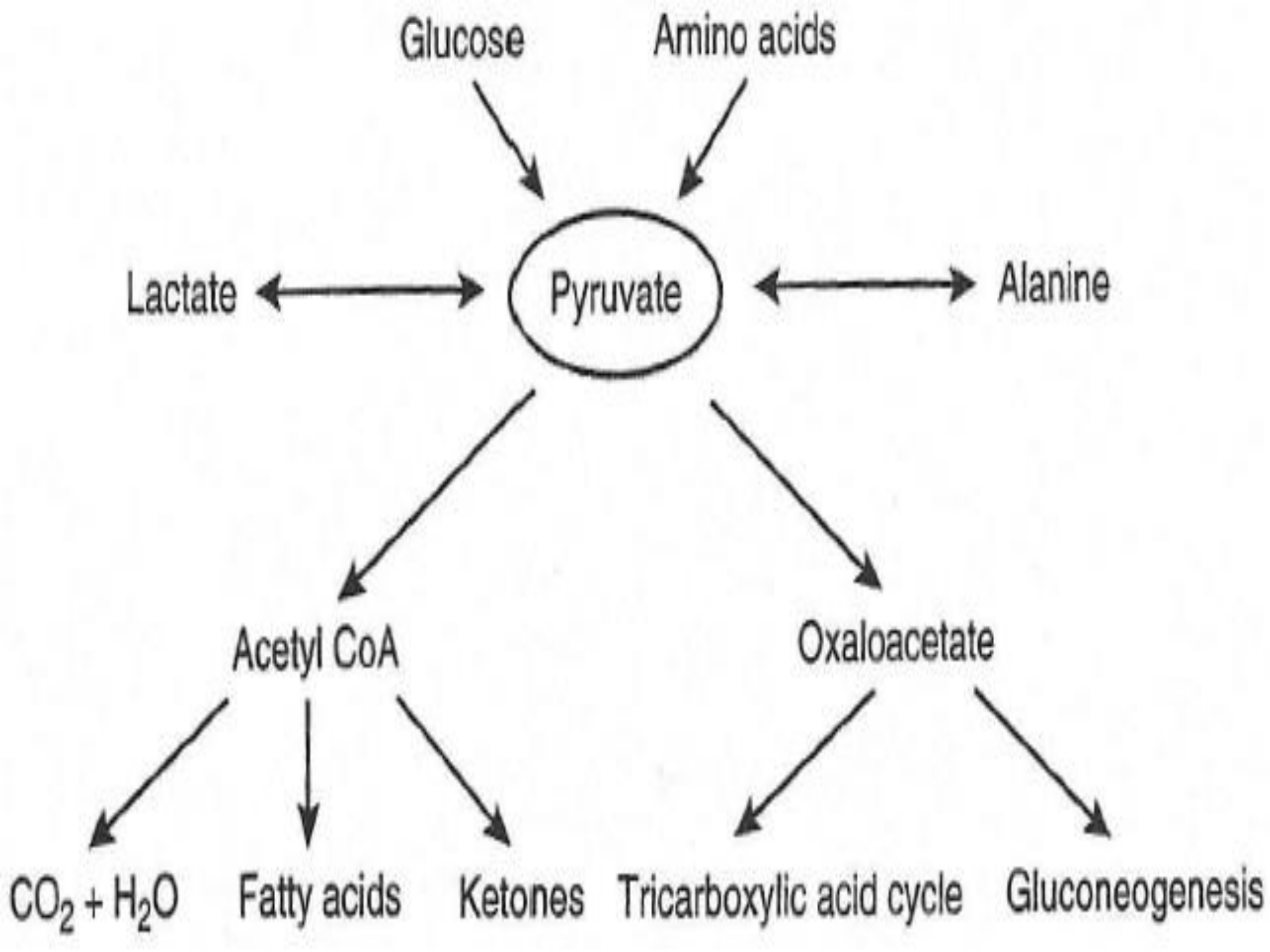
Pyruvate kinase

ADP

ATP

Pyruvate




Glucose
Amino acids
Lactate
Pyruvate
Alanine
Acetyl CoA
Oxaloacetate
$CO_2 + H_2O$
Fatty acids
Ketones
Tricarboxylic acid cycle
Gluconeogenesis

Primary Plant metabolites

- Primary metabolites comprise many different types of organic compounds, including, but not limited to, carbohydrates, lipids, proteins, and nucleic acids. They are found universally in the plant kingdom because they are the components or products of fundamental metabolic pathways or cycles such as glycolysis, the Krebs cycle, and the Calvin cycle.
- Because of the importance of these and other primary pathways in enabling a plant to synthesize, assimilate, and degrade organic compounds, primary metabolites are essential.

Primary Plant metabolites

- Examples of primary metabolites include energy rich fuel molecules, such as sucrose and starch, structural components such as cellulose, informational molecules such as DNA (deoxyribonucleic acid) and RNA (ribonucleic acid), and pigments, such as chlorophyll.
- In addition to having fundamental roles in plant growth and development, some primary metabolites are precursors (starting materials) for the synthesis of secondary metabolites.

Primary Plant metabolites

- Primary metabolites provides the small molecules that are employed as starting material for all of the important secondary metabolic pathways.
- The primary metabolites are , ofcourse, essential for health, as they form the basis of the human diet.
- However there are times when no clear division exists between primary and secondary metabolites.
- This is the case with fundamentally important primary metabolites known as carbohydrates and lipids.

Secondary Plant metabolites

- Plants also use other metabolic pathways to produce compounds that often have on readily apparent function.
- These are the secondary metabolites, and their pathways of synthesis and utilization constitute secondary metabolism.
- Secondary metabolites, are derived from primary metabolites but have a more limited distribution in the wonderful diversity of plants.
- The secondary compounds appear to play no direct role in a plant's primary metabolism, but apparently have an ecological function.

Secondary Plant metabolites

- Secondary metabolites are often coloured, fragrant, or flavourful compounds, and they may attract pollinators, help the plant adapt to environmental stressors, or serve as chemical defenses against microorganisms, insects and even other plants.
- Many thousands of secondary metabolites have been isolated from plants, and many of them have powerful physiological effects in humans and are used as medicines.
- It is only since the late twentieth century that secondary metabolites have been clearly recognized as having important functions in plants.

Secondary Plant metabolites

- Secondary metabolites can be broadly divided into three classes of compounds, according to their route of biosynthesis: terpenes, phenolics, and nitrogen containing compounds (alkaloids).
- However, these classes of compounds also include primary metabolites, so whether a compound is a primary or secondary metabolite is a distinction based not only on its chemical structure but also on its function and distribution within the plant kingdom.

Secondary Plant metabolites

- Terpenes are lipids synthesized from acetyl coenzyme A via the mevalonic acid pathway.
- Phenolic compounds are aromatic substances formed in variuos ways via the shikmic acid pathway or malonic acid pathway.
- The Nitrogen containing secondary products, such as alkaloids, are biosynthesized primarily from aromatic acids.

Secondary Plant metabolites

- **Alkaloids**

- Alkaloids are a large group of nitrogen-containing compounds, examples of which are known to occur in approximately 20 percent of all flowering plants.
- Closely related plant species often contain alkaloids of related chemical structure.
- The primary metabolites from which they are derived include amino acids such as tryptophan, tyrosine, and lysine.
- Alkaloid biosynthetic pathways can be long, and many alkaloids have correspondingly complex chemical structures.
- Alkaloids can block **ion** channels, inhibit **enzymes**, or interfere with neurotransmission, producing **hallucinations** , loss of coordination, convulsions, vomiting, and death.

Secondary Plant metabolites

- Alkaloids accumulate in plant organs such as leaves or fruits and are ingested by animals that consume those plant parts.
- Many alkaloids are extremely toxic, especially to mammals, and act as potent nerve poisons, enzyme inhibitors, or membrane transport inhibitors.
- In addition to being toxic, many alkaloids are also bitter or otherwise bad-tasting.
- Therefore, the presence of alkaloids and other toxic secondary metabolites can serve as a deterrent to animals, which learn to avoid eating such plants.

Medicinal Alkaloids

- Many potentially toxic plant-derived alkaloids have medicinal properties, as long as they are administered in carefully regulated doses.
- Alkaloids with important medicinal uses include morphine and codeine from the opium poppy and cocaine from the coca plant. These alkaloids act on the nervous system and are used as painkillers.
- Atropine, from the deadly nightshade plant, also acts on the nervous system and is used in anesthesia and ophthalmology.
- Vincristine and vinblastine from the periwinkle(vinca) plant are inhibitors of cell division and are used to treat cancers of the blood and lymphatic systems.

Medicinal Alkaloids

- Quinine from the bark of the cinchona tree is toxic to the Plasmodium parasite, which causes malaria, and has long been used in tropical and subtropical regions of the world.
- Other alkaloids are used as stimulants, including caffeine, present in coffee, tea, and cola plants (and the drinks derived from these plants), and nicotine, which is present in tobacco. Nicotine preparations are, paradoxically, also used as an aid in smoking cessation. Nicotine is also a very potent insecticide.
- For many years ground-up tobacco leaves were used for insect control, but this practice was superseded by the use of special formulations of nicotine. More recently the use of nicotine as an insecticide has been discouraged because of its toxicity to humans.

Terpenoids

- Terpenoids are derived from acetyl coenzyme A or from intermediates in glycolysis.
- They are classified by the number of five-carbon isoprenoid units they contain. Monoterpenes (containing two C5- units) are exemplified by the aromatic oils (such as menthol) contained in the leaves of members of the mint family.
- In addition to giving these plants their characteristic taste and fragrance, these aromatic oils have insect-repellent qualities.
  - Ex: The pyrethroids, which are monoterpene esters from the flowers of chrysanthemum and related species, are used commercially as insecticides.
  - They fatally affect the nervous systems of insects while being biodegradable and nontoxic to mammals, including humans.

Terpenoids

- Diterpenes are formed from four C5-units.

  Ex: Paclitaxel (commonly known by the brand name Taxol), a diterpene found in bark of the Pacific yew tree, is a potent inhibitor of cell division in animals. At the end of the twentieth century, paclitaxel was developed as a powerful new chemotherapeutic treatment for people with solid tumors, such as ovarian cancer patients.

- Triterpenoids (formed from six C_5- units) comprise the plant steroids, some of which act as plant hormones. These also can protect plants from insect attack, though their mode of action is quite different from that of the pyrethroids.
- Tetraterpenoids (eight C_5- units) include important pigments such as beta-carotene, which is a precursor of vitamin A, and lycopene, which gives tomatoes their red colour.

Terpenoids

- Rather than functioning in plant defense, the coloured pigments that accumulate in ripening fruits can serve as attractants to animals, which actually aid the plant in seed dispersal.
- The polyterpenes are polymers that may contain several thousand isoprenoid units. Rubber, a polyterpene in the latex of rubber trees that probably aids in wound healing in the plant, is also very important for the manufacture of tires and other products.

Phenolic Compounds

- Phenolic compounds are defined by the presence of one or more aromatic rings bearing a hydroxyl functional group. Many are synthesized from the amino acid phenylalanine.
- Simple phenolic compounds, such as salicylic acid, can be important in defense against fungal pathogens.
- Salicylic acid concentration increases in the leaves of certain plants in response to fungal attack and enables the plant to mount a complex defense response.
- Interestingly, aspirin, a derivative of salicylic acid, is routinely used in humans to reduce inflammation, pain, and fever.
- Some phenolics interfere with digestion, slow growth, block enzyme activity and cell division, or just taste awful.

Phenolic Compounds

- Lignin, a complex phenolic macromolecule, is laid down in plant secondary cell walls and is the main component of wood. It is a very important structural molecule in all woody plants, allowing them to achieve height, girth, and longevity.
- Lignin is also valuable for plant defense: Plant parts containing cells with lignified walls are much less palatable to insects and other animals than are non woody plants and are much less easily digested by fungal enzymes than plant parts that contain only cells with primary cellulose walls.

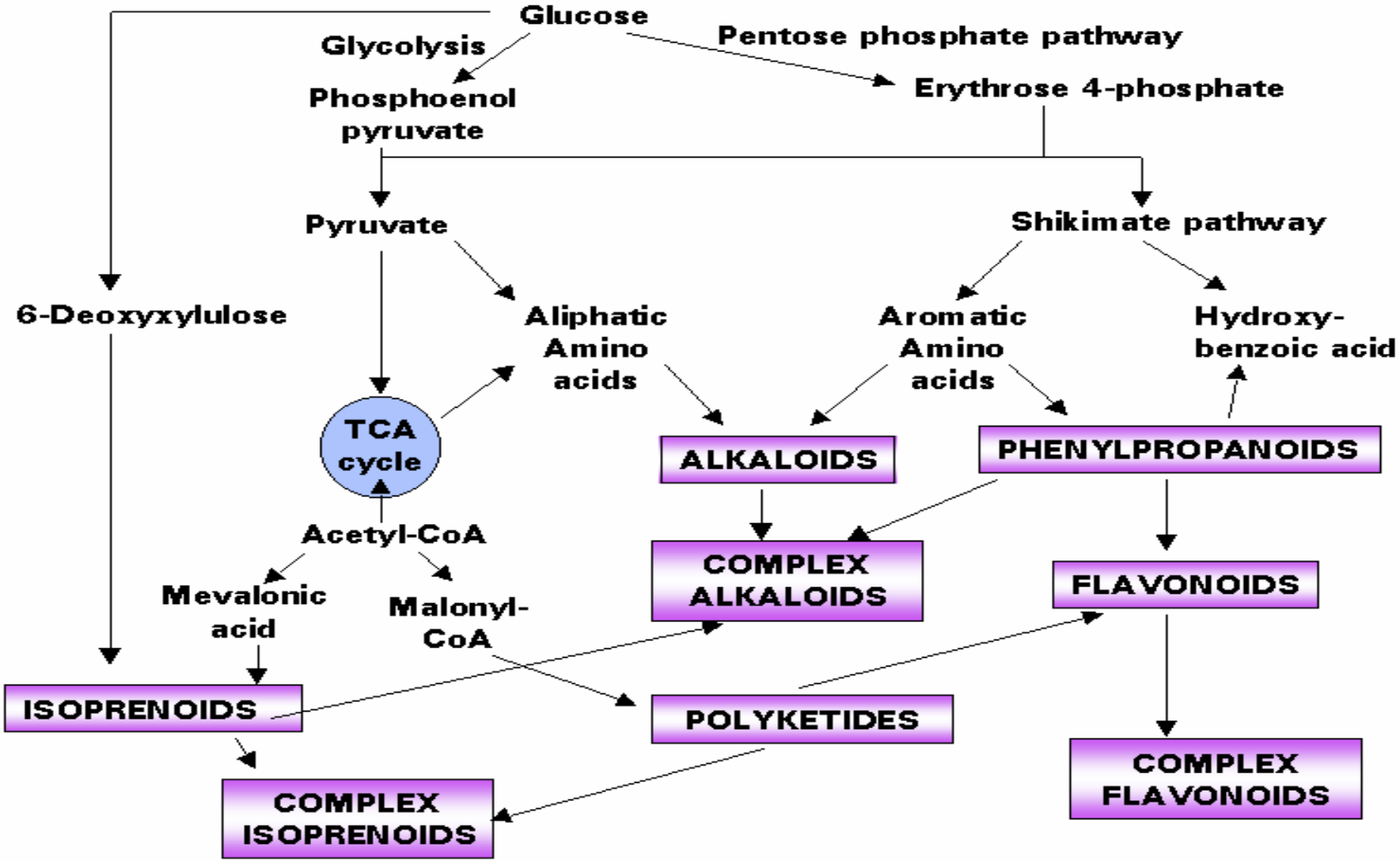
Phenolic Compounds

- Phenolics function as attractants. Anthocyanins and anthocyanidins are phenolic pigments that impart pink and purple colours to flowers and fruits. This pigmentation attracts insects and other animals that move between individual plants and accomplish pollination and fruit dispersal.
- Plants with red flowers attract birds and mammals because these animals possess the correct photoreceptors to see red pigments.
- Other phenolic compounds, called isoflavones, are synthesized rapidly in plants of the legume family when they are attacked by bacterial or fungal pathogens, and they have strong antimicrobial activity.

Class	Example Compounds	Example Sources	Some Effects and Uses
NITROGEN-CONTAINING			
Alkaloids	nicotine cocaine theobromine	tobacco coca plant chocolate (cocao)	interfere with neurotransmission, block enzyme action
NITROGEN-AND SULFUR-CONTAINING			
Glucosinolates	sinigrin	cabbage, relatives	
TERPENOIDS			
Monoterpenes	menthol linalool	mint and relatives, many plants	interfere with neurotransmission, block ion transport, anesthetic
Sesquiterpenes	parthenolid	Parthenium and relatives (Asteraceae)	contact dermatitis
Diterpenes	gossypol	cotton	block phosphorylation; toxic
Triterpenes, cardiac glycosides	digitogenin	Digitalis (foxglove)	stimulate heart muscle, alter ion transport
Tetraterpenoids	carotene	many plants	antioxidant; orange coloring
Terpene polymers	rubber	Hevea (rubber) trees, dandelion	gum up insects; airplane tires
Sterols	spinasterol	spinach	interfere with animal hormone action

Class	Example Compounds	Example Sources	Some Effects and Uses
PHENOLICS			
Phenolic acids	caffeic, chlorogenic	all plants	cause oxidative damage, browning in fruits and wine
Coumarins	umbelliferone	carrots, parsnip	cross-link DNA, block cell division
Lignans	podophyllin urushiol	mayapple poison ivy	cathartic, vomiting, allergic dermatitis
Flavonoids	anthocyanin, catechin	almost all plants	flower, leaf color; inhibit enzymes, anti- and pro-oxidants, estrogenic
Tannins	gallotannin, condensed tannin	oak, hemlock trees, birdsfoot trefoil, legumes	bind to proteins, enzymes, block digestion, antioxidants
Lignin	lignin	all land plants	structure, toughness, fiber

Secondary Metabolites are Derived from Primary Metabolites



Main Groups of Secondary Metabolites in Plants

- 29,000 terpenes - derived from the C5 precursor isopentenyl diphosphate (IPP)
- 12,000 alkaloids - derived from amino acids.
- 8,000 phenolics shikimate pathway or phenolics malonate/acetate pathway

Main Secondary metabolites

- Nitrogen containing:

  -Alkaloids (12,000)

  -Non protein amino acids (600)

  -Amines (100)

  -Cyanogenic glycosides (100)

  -Glucosinolates(100)

Main Secondary metabolites

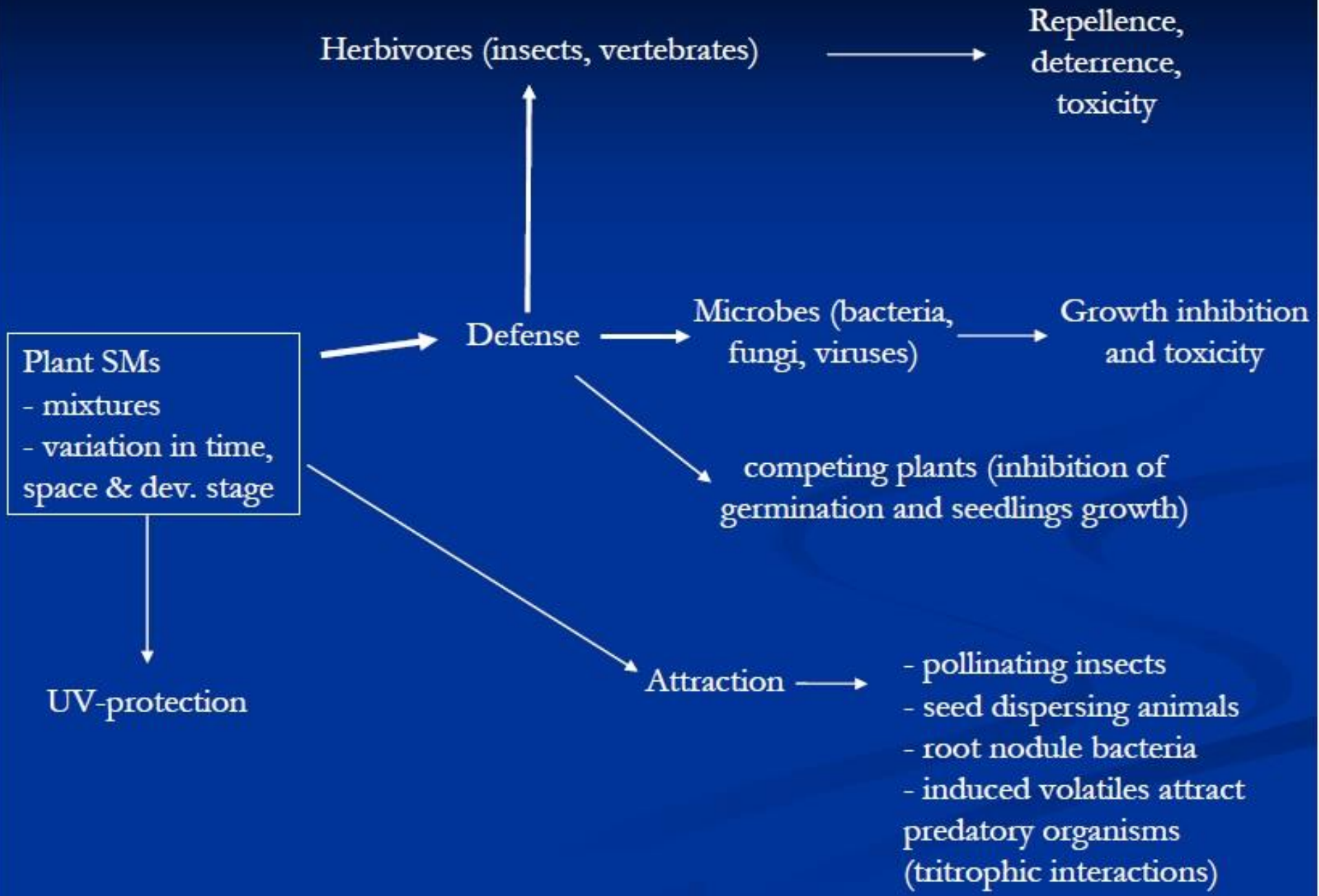
- Without nitrogen:
  -Terpenoids(29,000):
    mono-1000
    sesquiterpene-3000
    diterpenes-1000
    triterpenes, steroids, saponines-4,000
  -Phenolics (8,000):
    Flavonoids-2000
    Polyacetylens-1000
    Polyketides-750
    Phenylpropanoids-500

Secondary Metabolites biosynthesis in a cell

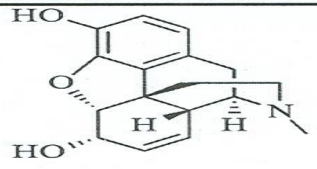
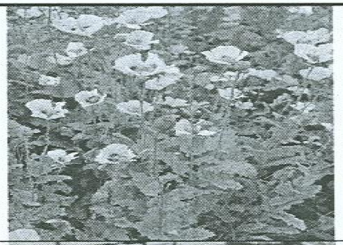
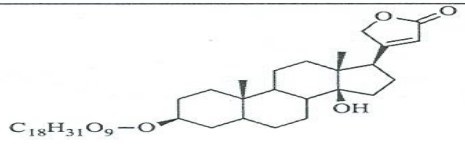
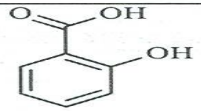
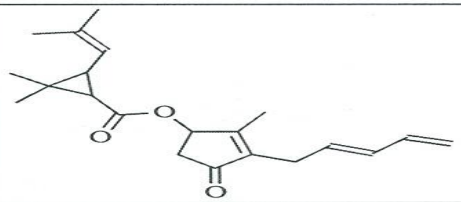
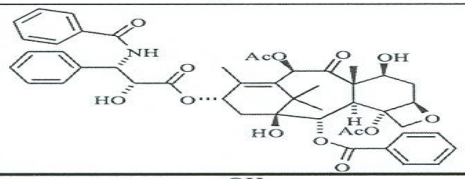
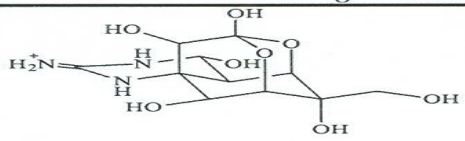
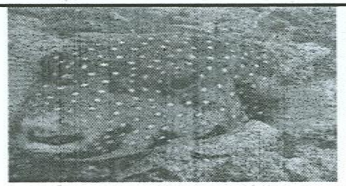
- Mostly in the Cytosol : hydrophilic compounds
- Chloroplasts : alkaloids (caffeine) and terpenoids (monoterpenes)
- Mitochondria : some amines, alkaloids
- Vesicles : alkaloids (protoberberines)
- Endoplamic reticulum: hydroxylaton steps, lipophilic compounds

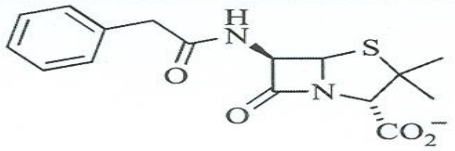
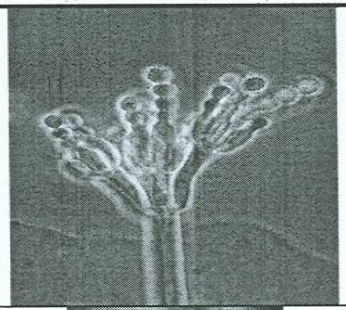
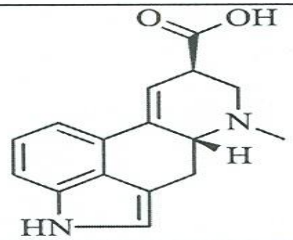
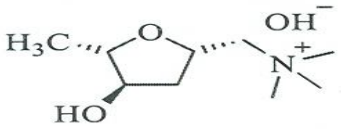
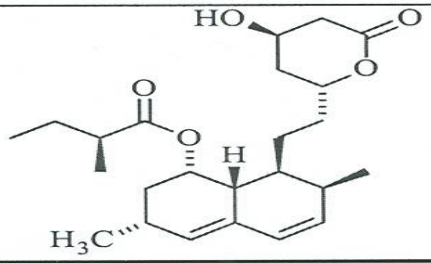
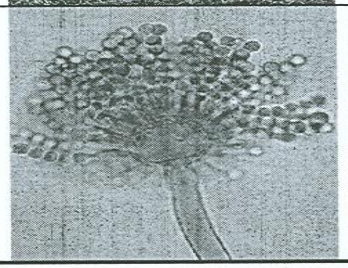
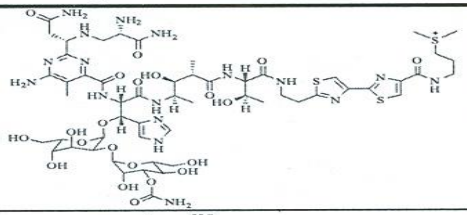
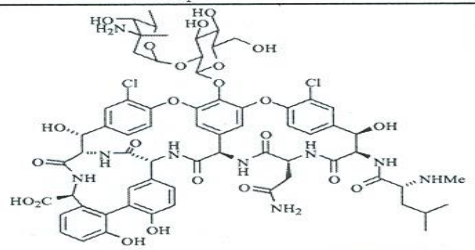
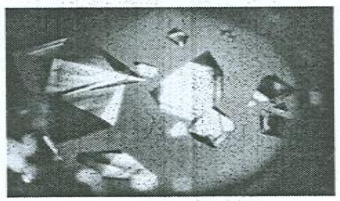
Water soluble compounds are usually stored in the vacuole

Function of Secondary Metabolites:



Examples of Secondary Metabolites

Name	Chemical Structure	Species		Uses
Morphine		*Papaver somniferum* (Poppy)		Pain control
Digitoxin		*Digitalis purpurea* (Foxglove)		Treatment of congestive heart failure and to control ventricular rate
Salicylic acid		*Salix sp.* (Willow)		Analgesic Anti-inflammatory
Pyrethrin I		*Compositae* family (Including Chrysanthemum and *Pyrethrum pulchrum*)		Insecticide
Taxol		*Taxus brevifolia* (Pacific Yew)		Anticancer agent
Tetrodotoxin		*Lagocephalus lagocephalus* (Puffer fish)		Poison

Penicillin		*Penicillium sp.*		Antibiotic
Lysergic acid		*Claviceps sp.* (Ergot)		Poison
Muscarine		*Amanita muscaria* (Fly agaric)		Poison Parasympathomi-metic
Lovastatin		*Aspergillus terreus*		Lowering of Cholesterol
Bleomycin (bleocin)		*Streptomyces verticillus*		Antitumour antibiotic
Vancomycin		*Nocardia orientalis* (actinomycete formerly known as *Streptomyces orientalis*)		Antibiotic